

# Orthogonal Bases and Gram-Schmidt(标准正交基和施密特正交化过程)

## Lecture 16

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# Orthogonal Bases and Gram-Schmidt

- 1 Introduction
- 2 Linear Combinations
- 3 The Gram-Schmidt Process
- 4 Homework Assignment 16

# Orthogonal Vectors

- In an orthogonal basis, every vector is perpendicular to every other vector.
- The coordinate axes are mutually orthogonal. That is just about optimal, and the one possible improvement is easy: Divide each vector by its length, to make it a unit vector.

## Definition

The vectors  $q_1, q_2, \dots, q_n$  are orthonormal if

$$q_i^T q_j = \begin{cases} 0 & \text{whenever } i \neq j, \text{ giving the orthogonality;} \\ 1 & \text{whenever } i = j, \text{ giving the normalization.} \end{cases}$$

A matrix with orthonormal columns will be called  $Q$ .

# Introduction

If we have a subspace of  $\mathbb{R}^n$ , the standard vectors  $e_i$  might not lie in that subspace. But the subspace always has an orthonormal basis, and it can be constructed in a simple way out of any basis whatsoever. This construction, which **converts a skewed set of axes into a perpendicular set**, is known as **Gram-Schmidt orthogonalization**. To summarize, the three topics basic to this section are:

1. The definition and properties of orthogonal matrices  $Q$ .
2. The solution of  $Qx = b$ , either  $n$  by  $n$  or rectangular (least-squares).
3. The Gram-Schmidt process and its interpretation as a new factorization  $A = QR$ .

# Orthogonal Matrices

## Proposition

*If  $Q$  (square or rectangular) has orthonormal columns, then  $Q^T Q = I$ :*

$$\begin{bmatrix} q_1^T \\ q_2^T \\ \dots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix} = I$$

*An orthogonal matrix is a square matrix with orthonormal columns. Then  $Q^T = Q^{-1}$ . For square orthogonal matrices, the transpose is the inverse.*

Note that  $Q^T Q = I$  even if  $Q$  is rectangular. But then  $Q^T$  is only a left-inverse.

# Examples

## Example

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad Q^T = Q^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}.$$

$Q$  rotates every vector through the angle  $\theta$ , and  $Q^T$  rotates it back through  $-\theta$ . The columns are clearly orthogonal, and they are orthonormal because  $\sin^2 \theta + \cos^2 \theta = 1$ . The matrix  $Q^T$  is just as much an orthogonal matrix as  $Q$ .

## Example

Any permutation matrix  $P$  is an orthogonal matrix. Geometrically, an orthogonal  $Q$  is the product of a rotation and a reflection.

# Length Preserving

There does remain one property that is shared by rotations and reflections, and in fact by every orthogonal matrix.

## Proposition

*Multiplication by any  $Q$  preserves lengths:*

$$\|Qx\| = \|x\| \quad \text{for every vector } x$$

Remarks:

- This property is not shared by projections, which are not orthogonal or even invertible.
- Projections reduce the length of a vector, whereas orthogonal matrices preserve lengths.

# Linear Combinations

## Write $b$ as a combination

$$b = x_1 q_1 + x_2 q_2 + \cdots + x_n q_n,$$

to find  $x_i$ , we multiply both sides of the equation by  $q_i^T$ .

Remarks:

- Every vector  $b$  is the sum of its one-dimensional projections onto the lines through the  $q$ 's.
- The rows of a square matrix are orthonormal whenever the columns are. Example:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

# Rectangular matrices with Orthogonal Columns

If  $Q$  has orthonormal columns, the least-squares problem becomes easy: rectangular system with no solution for most  $b$  and the projection matrix is  $P = QQ^T$ .

$Qx = b$       rectangular system with no solution for most  $b$

$Q^T Q \hat{x} = Q^T b$       normal equations for the best  $\hat{x}$  – in which  $Q^T Q = I$ .

$\hat{x} = Q^T b$        $\hat{x}_i$  is  $q_i^T b$ .

$p = Q\hat{x}$       the projection of  $b$  is  $(q_1^T b)q_1 + \cdots + (q_n^T b)q_n$ .

$p = QQ^T b$       the projection matrix is  $P = QQ^T$ .

## Remarks

The last formulas are like  $p = A\hat{x}$  and  $P = A(AA^T)^{-1}A^T$ . When the columns are orthonormal, the “cross product matrix”  $A^TA$  becomes  $Q^TQ = I$ . The projections onto the axes are uncoupled, and  $p$  is the sum

$$p = (q_1^T b)q_1 + \cdots + (q_n^T b)q_n.$$

# Examples

## Example

Project a point  $b = (x, y, z)$  onto the  $x$ - $y$  plane.

Remark:

**Projection onto a plane = sum of projections onto orthonormal  $q_1$  and  $q_2$ .**

# Example

## Example

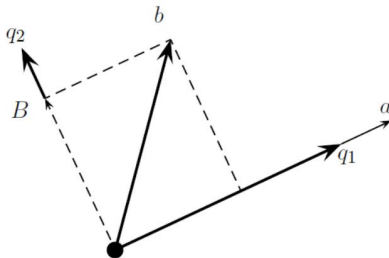
**Example 4** When the measurement times average to zero, fitting a straight line leads to orthogonal columns. Take  $t_1 = -3$ ,  $t_2 = 0$ , and  $t_3 = 3$ . Then the attempt to fit  $y = C + Dt$  leads to three equations in two unknowns:

$$\begin{cases} C + Dt_1 = y_1 \\ C + Dt_2 = y_2 \\ C + Dt_3 = y_3 \end{cases} \quad \text{or} \quad \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

The columns  $(1, 1, 1)$  and  $(-3, 0, 3)$  are orthogonal. We can project  $y$  separately onto each column, and the best coefficients  $\hat{C}$  and  $\hat{D}$  can be found separately.

# The Gram-Schmidt Process

- Suppose you are given three independent vectors  $a, b, c$ . If they are orthonormal, life is easy. If they are not orthonormal, we need to propose a way to make them orthonormal.
- The idea is to subtract from every new vector its components in the directions that are already settled.



**Figure 3.10:** The  $q_i$  component of  $b$  is removed;  $a$  and  $B$  normalized to  $q_1$  and  $q_2$ .

# The Gram-Schmidt Process

- The Gram-Schmidt process starts with independent vectors  $a_1, a_2, \dots, a_n$  and ends with orthonormal vectors  $q_1, q_2, \dots, q_n$ .
- At step  $j$  it subtracts from  $a_j$  its components in the directions  $q_1, \dots, q_{j-1}$  that are already settled:

$$A_j = a_j - (q_1^T a_j)q_1 - \dots - (q_{j-1}^T a_j)q_{j-1}.$$

- Then  $q_j$  is the unit vector

$$q_j = \frac{A_j}{\|A_j\|}.$$

# Example: Gram-Schmidt

**Example 5. Gram-Schmidt** Suppose the independent vectors are  $a, b, c$ :

$$a = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

$$B = b - (q_1^T b)q_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

$$\begin{aligned} C &= c - (q_1^T c)q_1 - (q_2^T c)q_2 \\ &= \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} - \sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \end{aligned}$$

**Orthonormal basis**  $Q = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}.$

# The Factorization $A = QR$

- Every  $m$  by  $n$  matrix with independent columns can be factored into  $A = QR$ .
- The columns of  $Q$  are orthonormal, and  $R$  is upper triangular and invertible.
- When  $m = n$  and all matrices are square,  $Q$  becomes an orthogonal matrix.

Remark on the calculations:

- It is easier to compute the orthogonal  $Q$ , without forcing their lengths to equal one.

## QR:Example

We started with a matrix  $A$ , whose columns were  $a, b, c$ . We ended with a matrix  $Q$ , whose columns are  $q_1, q_2, q_3$ . The  $QR$  factorization is as follows:

$$QR = \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix}$$

From example 5, we deduce that:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \sqrt{2} \\ 0 & \frac{1}{\sqrt{2}} & \sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} = QR.$$

- You see the lengths of  $a, B, C$  on the diagonal of  $R$ .
- The orthonormal vectors  $q_1, q_2, q_3$ , which are the whole object of orthogonalization, are in the first factor  $Q$ .

## $QR$ factorization

Maybe  $QR$  is not as beautiful to the theory as  $LU$  (because of the square roots). Both factorizations are vitally important to the theory of linear algebra, and absolutely central to the calculations. If  $LU$  is Hertz, then  $QR$  is Avis.

### Theorem

*Every  $m$  by  $n$  matrix with independent columns can be factored into  $A = QR$ . The columns of  $Q$  are orthonormal, and  $R$  is upper triangular and invertible. When  $m = n$  and all matrices are square,  $Q$  becomes an orthogonal matrix.*

# Function Spaces and Fourier Series

1. Hilbert Space.
2. Lengths and Inner Products.
3. Fourier Series.
4. Gram-Schmidt for Functions.
5. Best Straight Line.

# Homework Assignment 16

3.4: 1, 4, 5, 6, 13, 15, 17.